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Ultra Safe Capsule – Generation 1

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1. Executive Overview

Ultra Safe Capsule – Generation 1 is a large-scale, ultra-stable aerial platform engineered with a triple-layer architecture, hybrid buoyancy-lift system, and multi-redundant safety design. The system combines helium static lift with low-RPM electric propulsion to achieve maximum safety, ultra-low noise, and extreme hover stability.

The project is engineered with a "Safety-First Over-Engineering" philosophy and is optimized for controlled hover, emergency survivability, and structural resilience.

2. General Dimensions & Geometry

Overall Width: 70 meters Overall Length: ~70 meters Overall Height: 32 meters

Design Profile: Wide-body, low vertical profile

Design Rationale: - Low center of gravity - Reduced overturning moment -
Improved crosswind stability - Aerodynamic drag moderation

3. Triple-Layer Architecture

Layer 1 – Core Lift Platform

- Primary structural frame
- Main propulsion integration
- Landing system integration
- Load distribution structure

Layer 2 – Helium Chamber

- Segmented helium gas cells
- Structural containment membrane
- Pressure management system
- Volume: ~107,100 cubic meters

Layer 3 – Outer Safety Capsule

- Independent safety shell
 - Secondary propulsion ring
 - Dedicated parachute system
 - Protected operational zone
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4. Propulsion Architecture

Primary Lift System (Layer 1)

- 4 × 20-meter diameter fans
- Low RPM, high torque configuration
- Designed to compensate effective load (~8 tons)
- Silent operational profile

Stability & Maneuvering Ring (Layer 3)

- 9 × 9-meter concealed fans
 - Directional stabilization
 - Yaw and lateral correction
 - Hover precision control
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5. Structural Materials

Primary Structure Composition: - 80% Carbon Fiber Reinforced Polymer (CFRP) - 20% High-strength metal alloys (mechanical subsystems)

Engineering Approach: - Advanced bonding instead of heavy fastener systems - Weight optimization through composite shell integration - Structural redundancy in high-load nodes

Estimated Dry Weight: ~120 metric tons

6. Buoyancy & Lift Balance

Helium Volume: ~107,100 m³ Estimated Static Lift: ~112 metric tons Total Structural Weight: ~120 metric tons

Effective Load in Air: $120 - 112 = \sim 8$ metric tons

Implication: The propulsion system compensates only ~8 tons during hover, enabling: - Extremely low energy demand - Reduced mechanical stress - High operational safety margin

7. Landing System

- 16 mechanical/hydraulic landing legs
- Each leg width: 3 meters

- Leg structural height: 1 meter
 - Even ground load distribution
 - Shock absorption capability
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8. Multi-Redundant Safety System

Main Hull Parachutes

- 16 distributed parachutes

Capsule Parachutes

- 16 dedicated parachutes for Layer 3

Redundancy Philosophy

- Multi-trigger deployment system
- Independent capsule survivability
- Layer-isolated emergency response

Total Parachute Units: 32

9. Operational Advantages

- Ultra-stable hover platform
 - Extremely low acoustic signature
 - High survivability architecture
 - Balanced lift-to-weight configuration
 - Over-engineered safety margins
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10. Engineering Status – Generation 1

- Geometry finalized
- Lift-to-mass balance validated
- Structural composition defined
- Propulsion architecture confirmed
- Safety redundancy established

Ultra Safe Capsule – Generation 1 is structurally stabilized at the conceptual engineering phase and prepared for detailed internal systems design and investor presentation.

Founder Amirbahador ATAI

Ultra Safe Capsule Program Generation 1 – Master Baseline Document
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